

PYTHON PROJECT

STUDENT MANAGEMENT SYSTEM

A Ready-to-Run College Project

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Technology: Python 3

Project Type: Console-Based Application

College Project Submission

1. Project Introduction

The Student Management System is a simple Python-based application designed to manage student records. It allows the user to add, view, search, update and delete student information through a menu-driven interface. The project is suitable for demonstrating Python fundamentals such as variables, functions, loops, conditional statements, lists/dictionaries and file handling.

2. Objectives

- Create a simple and user-friendly student record system.
- Store student details in a local file.
- Provide options to add, display, search, update and delete records.
- Demonstrate practical use of Python programming concepts.

3. Main Features

1. Add Student
2. View All Students
3. Search Student by Roll Number
4. Update Student Details
5. Delete Student Record
6. Data Storage using JSON file
7. Exit

4. Software Requirements

- Python 3.x
- Windows/Linux/macOS
- Any Python editor such as IDLE, VS Code or PyCharm
- No external Python libraries are required.

5. Hardware Requirements

- Basic computer/laptop
- Minimum 2 GB RAM recommended
- Keyboard and display

6. Complete Python Source Code

Save the following code as **student_management.py** and run it using Python 3.

```
import json
import os

FILE_NAME = "students.json"

def load_students():
    if os.path.exists(FILE_NAME):
        try:
            with open(FILE_NAME, "r") as file:
                return json.load(file)
        except:
            return []
    return []

def save_students(students):
    with open(FILE_NAME, "w") as file:
        json.dump(students, file, indent=4)

def add_student(students):
    roll = input("Enter Roll Number: ").strip()

    for student in students:
        if student["roll"] == roll:
            print("Student with this roll number already exists.")
            return

    name = input("Enter Student Name: ").strip()
    course = input("Enter Course: ").strip()
    marks = input("Enter Marks: ").strip()

    students.append({
        "roll": roll,
        "name": name,
        "course": course,
        "marks": marks
    })
    save_students(students)
    print("Student added successfully.")

def view_students(students):
    if not students:
        print("No student records found.")
        return

    print("\n--- ALL STUDENTS ---")
    for s in students:
        print(f"Roll: {s['roll']} | Name: {s['name']} | "
              f"Course: {s['course']} | Marks: {s['marks']}")

def search_student(students):
    roll = input("Enter Roll Number to search: ").strip()

    for s in students:
        if s["roll"] == roll:
            print("\nStudent Found")
            print("Name      :", s["name"])
            print("Course   :", s["course"])
            print("Marks    :", s["marks"])
            return

    print("Student not found.")

def update_student(students):
    roll = input("Enter Roll Number to update: ").strip()

    for s in students:
        if s["roll"] == roll:
            print("Press Enter to keep the old value.")
            name = input(f"Name [{s['name']}]: ").strip()
            course = input(f"Course [{s['course']}]: ").strip()
            marks = input(f"Marks [{s['marks']}]: ").strip()

            if name:
```

```

        s["name"] = name
    if course:
        s["course"] = course
    if marks:
        s["marks"] = marks

    save_students(students)
    print("Student updated successfully.")
    return

print("Student not found.")

def delete_student(students):
    roll = input("Enter Roll Number to delete: ").strip()

    for i, s in enumerate(students):
        if s["roll"] == roll:
            students.pop(i)
            save_students(students)
            print("Student deleted successfully.")
            return

    print("Student not found.")

def main():
    students = load_students()

    while True:
        print("\n===== STUDENT MANAGEMENT SYSTEM =====")
        print("1. Add Student")
        print("2. View All Students")
        print("3. Search Student")
        print("4. Update Student")
        print("5. Delete Student")
        print("6. Exit")

        choice = input("Enter your choice: ").strip()

        if choice == "1":
            add_student(students)
        elif choice == "2":
            view_students(students)
        elif choice == "3":
            search_student(students)
        elif choice == "4":
            update_student(students)
        elif choice == "5":
            delete_student(students)
        elif choice == "6":
            print("Thank you for using the system.")
            break
        else:
            print("Invalid choice. Please try again.")

if __name__ == "__main__":
    main()

```

7. How to Run the Project

Step 1: Install Python 3 on your computer.

Step 2: Open VS Code, IDLE or another Python editor.

Step 3: Create a file named **student_management.py**.

Step 4: Paste the complete code into the file.

Step 5: Run the program.

Step 6: A **students.json** file will be created automatically when records are saved.

8. Sample Output

```
===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit

Enter your choice: 1
Enter Roll Number: 101
Enter Student Name: Shivam
Enter Course: BCA
Enter Marks: 78
Student added successfully.

Enter your choice: 2

--- ALL STUDENTS ---
Roll: 101 | Name: Shivam | Course: BCA | Marks: 78
```

9. Concepts Used

- Variables and data types
- Functions
- if-elif-else conditions
- while and for loops
- Lists and dictionaries
- JSON file handling
- Exception handling
- Modular program structure

10. Advantages

The project is easy to understand, requires no external database, stores data permanently in a JSON file, and demonstrates several important Python concepts in one practical application.

11. Future Scope

The project can be upgraded with a graphical user interface using Tkinter, a MySQL/SQLite database, login authentication, attendance management, fee management, report generation and an export-to-Excel feature.

12. Conclusion

The Student Management System is a practical beginner-to-intermediate Python project. It demonstrates how Python can be used to build a useful real-world application for managing student records. The project is suitable for a college demonstration and can be further enhanced with a GUI and database.

Project Status: Ready to Run